

mb-pc40s-2026-jun

Typst conversion review. OCR content and mathematical accuracy require editorial review.

Question 1

Jack rolled a water bottle from one edge of his desk to the other. The desk measures 60 cm from edge to edge. The water bottle has a diameter of 7 cm . Determine the angle that the water bottle rotated, in degrees.

Solution

$$s = \theta r$$

$$\theta = \frac{s}{r}$$

$$\theta = \frac{60}{3.5}$$

$$\theta = 17.142857...$$

$$\theta = (17.142857...) \left(\frac{180^\circ}{\pi} \right)$$

$$\theta = 982.213363...$$

$$\theta = 982.213^\circ$$

1 mark for substitution

1 mark for conversion

2 marks

Question 2

There are 15 dogs and 12 cats in an animal shelter. Determine the number of ways that three dogs and two cats can be selected if Scout, one of the dogs, must be selected.

Solution

$${}_1C_1 \cdot {}_{14}C_2 \cdot {}_{12}C_2$$
$$6006$$

$$1 \text{ mark for } {}_{14}C_2$$

$$\frac{1}{2} \text{ mark for } {}_{12}C_2$$

$$\frac{1}{2} \text{ mark for product of combinations}$$

2 marks

Note:

${}_1C_1$ does not need to be shown.

Question 3

Emily wants to save money to buy a car. She invests \$180 per month at an annual interest rate of 4.5%, compounded monthly.

Determine, algebraically, the number of monthly investments she will need to make to obtain at least \$15000. Express the final answer as a whole number.

Use the formula: $FV = \frac{R[(1+i)^n - 1]}{i}$

where FV = the future value

R = the investment amount each period

$i = \left[\frac{\text{the annual interest rate (as a decimal)}}{\text{the number of compounding periods per year}} \right]$

n = the number of investments

Solution

$15000 = \frac{180 \left[\left(1 + \frac{0.045}{12} \right)^n - 1 \right]}{\frac{0.045}{12}}$	$\frac{1}{2}$ mark for substitution
$15000 = \frac{180[(1 + 0.00375)^n - 1]}{0.00375}$	$\frac{1}{2}$ mark for simplification
$\log(1.3125) = \log(1.00375)^n$	$\frac{1}{2}$ mark for applying logarithms
$\log(1.3125) = n \log(1.00375)$	1 mark for power law
$n = \frac{\log(1.3125)}{\log(1.00375)}$	$\frac{1}{2}$ mark for evaluating quotient of logarithms
$n = 72.651539...$	3 marks
$\therefore 73 \text{ monthly investments are needed}$	

Question 4

Determine and simplify the 4th term in the binomial expansion of $\left(3x - \frac{2}{x^2}\right)^6$.

Solution

$$\begin{aligned}t_4 &= {}_6C_3(3x)^3\left(-\frac{2}{x^2}\right)^3 \\&= 20(27x^3)\left(\frac{-8}{x^6}\right) \\&= \frac{-4320}{x^3}\end{aligned}$$

2 marks (1 mark for ${}_6C_3$; $\frac{1}{2}$ mark for each consistent factor)

1 mark for simplification ($\frac{1}{2}$ mark for coefficient, $\frac{1}{2}$ mark for exponent)

3 marks

Question 5

Solve, algebraically, over the interval $[0, 2\pi]$.

$$4 \cos^2 x - 3 \cos x - 1 = 0$$

Solution

$$4 \cos^2 x - 3 \cos x - 1 = 0$$

$$(4 \cos x + 1)(\cos x - 1) = 0$$

$$\cos x = -\frac{1}{4} \quad \cos x = 1 \quad 1 \text{ mark for solving for } \cos x (1/2 \text{ mark for each branch})$$

$$x_r = 1.318116...$$

$$x = 1.823, 4.460 \quad x = 0, 2\pi \quad 2 \text{ marks for solving for } x (1/2 \text{ mark for each value})$$

3 marks

Question 6

State an equation for a rational function, $g(x)$, whose graph has a vertical asymptote at $x = 7$.

$g(x) =$

Solution

$$g(x) = \frac{1}{x - 7}$$

1 mark

Note:

Other equations are possible.

Question 7

State an equation of a radical function, $f(x)$, with a domain of $x \leq 0$ and a range of $y \geq 1$.

$f(x) =$

—

Solution

$f(x) = \sqrt{-x} + 1$ 1 mark for a radical function with a domain of $x \leq 0$

1 mark for a radical function with a range of $y \geq 1$

2 marks

Note:

Other equations are possible.

Question 8

Prove the identity for all permissible values of x .

$$\frac{\cos x + \sin^2 x \sec x}{\sin x} = \sec x \csc x$$

Left-Hand Side	Right-Hand Side

Solution

Method 1

Left-Hand Side	Right-Hand Side
$\frac{\cos x + \sin^2 x \sec x}{\sin x}$ $\frac{\cos x + \frac{\sin^2 x}{\cos x}}{\sin x}$ $\frac{\cos^2 x + \sin^2 x}{\cos x}$ $\frac{\sin x}{\frac{1}{\cos x}}$ $\sin x \cdot \frac{1}{\cos x}$ $\sec x \csc x$	$\sec x \csc x$

1 mark for correct substitution of appropriate identities

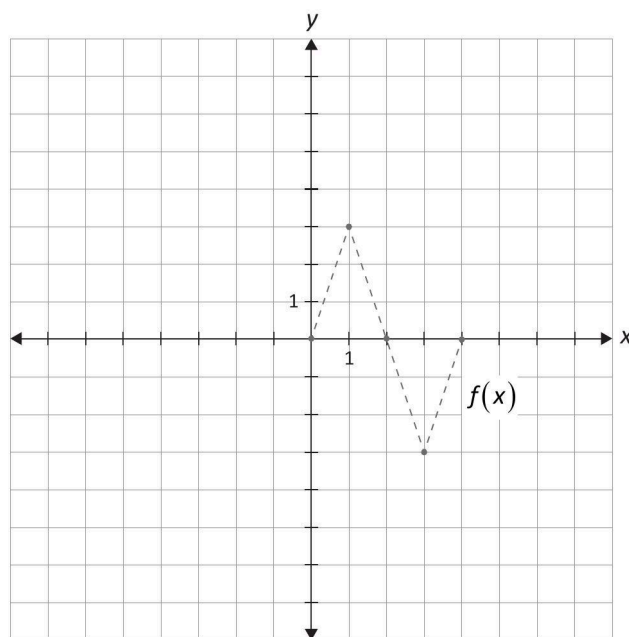
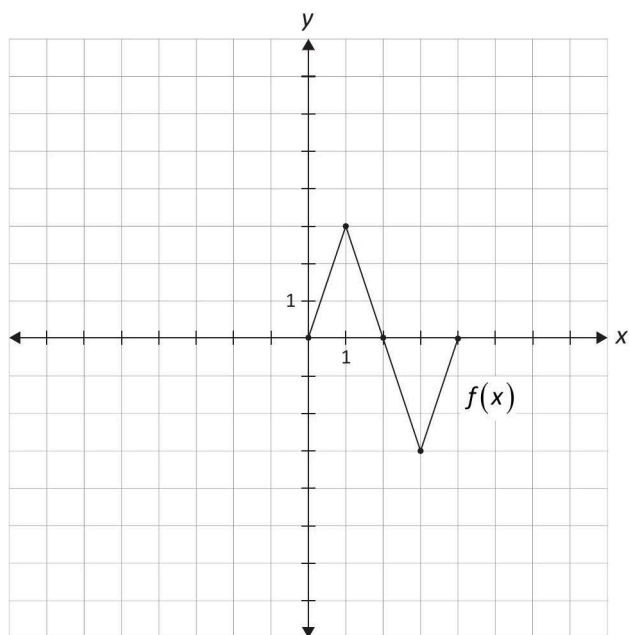
1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Question 9

Given the graph of, $y = f(x)$, sketch the graph of $y = 2 | f(-x) |$.

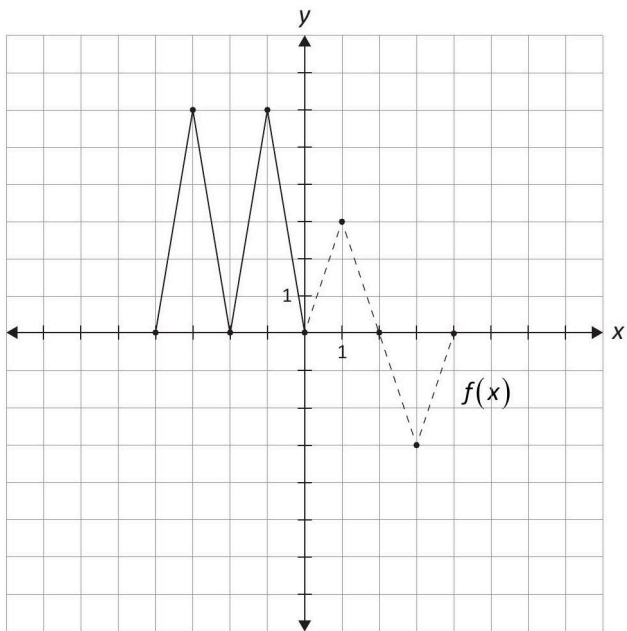


The graph of
 $f(x)$ has already
been drawn for
your reference.

No marks will be
awarded for the
graph of $f(x)$.

Solution

Solution



1 mark for horizontal reflection

1 mark for absolute value

1 mark for vertical stretch

3 marks

Question 10

Solve, algebraically.

$$\log(x - 1) + \log(x + 2) = 1$$

Solution

$$\begin{aligned}\log[(x - 1)(x + 2)] &= 1 && 1 \text{ mark for product law} \\(x - 1)(x + 2) &= 10^1 && 1 \text{ mark for exponential form} \\x^2 + x - 2 &= 10 \\x^2 + x - 12 &= 0 \\(x + 4)(x - 3) &= 0\end{aligned}$$

$\frac{1}{2}$ mark for the permissible value of x

$\frac{1}{2}$ mark for showing the rejection of the extraneous root

3 marks

Question 11

Justify that there are only two negative terms in the expansion of $(-2x + 5y)^4$.

Solution

$$t_1 = {}_4C_0(-2x)^4(5y)^0 \rightarrow +$$

$$t_2 = {}_4C_1(-2x)^3(5y)^1 \rightarrow -$$

$$t_3 = {}_4C_2(-2x)^2(5y)^2 \rightarrow +$$

$$t_4 = {}_4C_3(-2x)^1(5y)^3 \rightarrow -$$

$$t_5 = {}_4C_4(-2x)^0(5y)^4 \rightarrow +$$

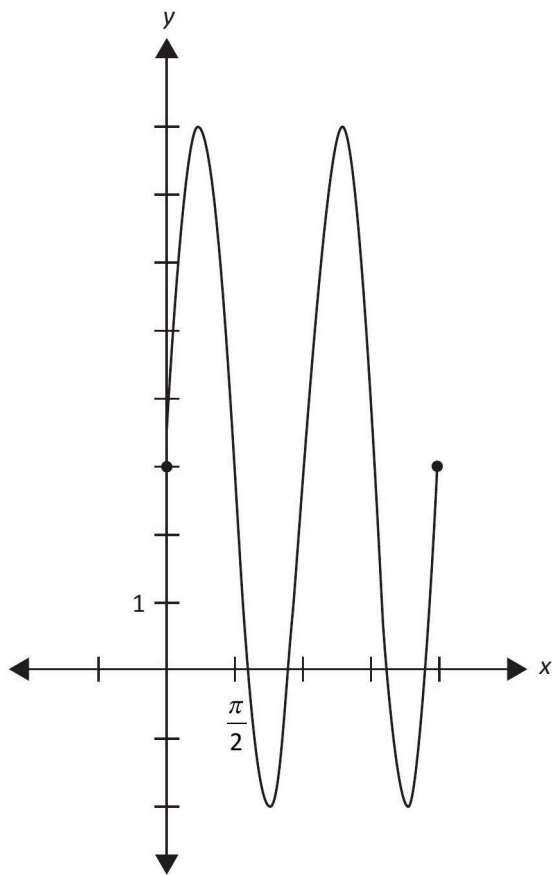
$\therefore$ The term is negative only when $(-2x)$ has an odd exponent, which occurs twice.

1 mark

Question 12

The graph of $y = 5 \sin(2x) + 3$ below can be used to solve the equation $0 = 5 \sin(2x) + 3$.

State how many solutions there are to the equation $0 = 5 \sin(2x) + 3$ over the interval $[0, 2\pi]$.



Solution

There are four solutions.

1 mark

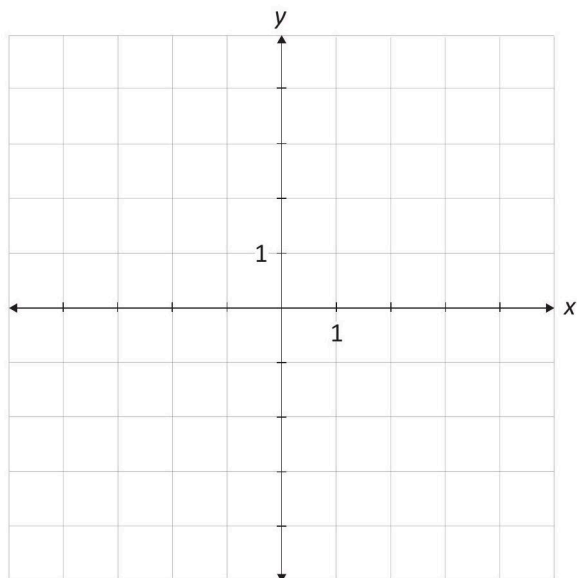
Question 13

Given the functions, $f(x) = x^2 - 1$ and $g(x) = x + 1$,

a) state the equation of $h(x) = \frac{f(x)}{g(x)}$.

$h(x) =$

b) sketch the graph of $h(x)$.



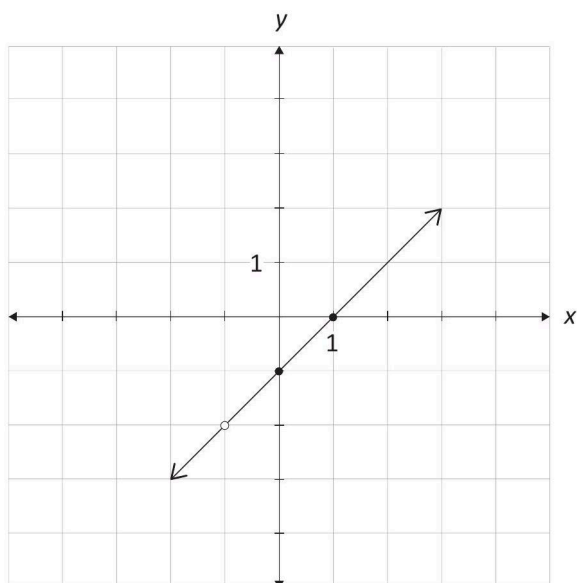
Solution

a)

$$\begin{aligned} h(x) &= \frac{x^2 - 1}{x + 1} \\ &= \frac{(x + 1)(x - 1)}{(x + 1)} \\ &= x - 1, x \neq -1 \end{aligned}$$

1 mark

b)



1 mark for shape of graph

consistent with a)

1 mark for point of discontinuity

(hole) at $x = -1$

2 marks

Note:

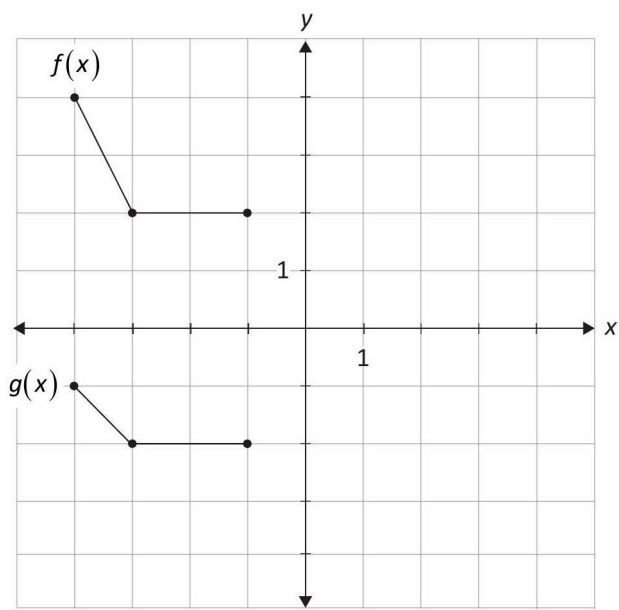
Deduct a maximum of 1 mark for the concept error of not restricting domain.

Deduct $\frac{1}{2}$ mark for procedural error (not stating domain of simplified function in part a)

if graph shows the correct domain.

Question 14

State an equation for $g(x)$, in terms of $f(x)$.



$g(x) =$

Solution

$g(x) = \frac{1}{2}f(x) - 3$

1 mark for vertical compression

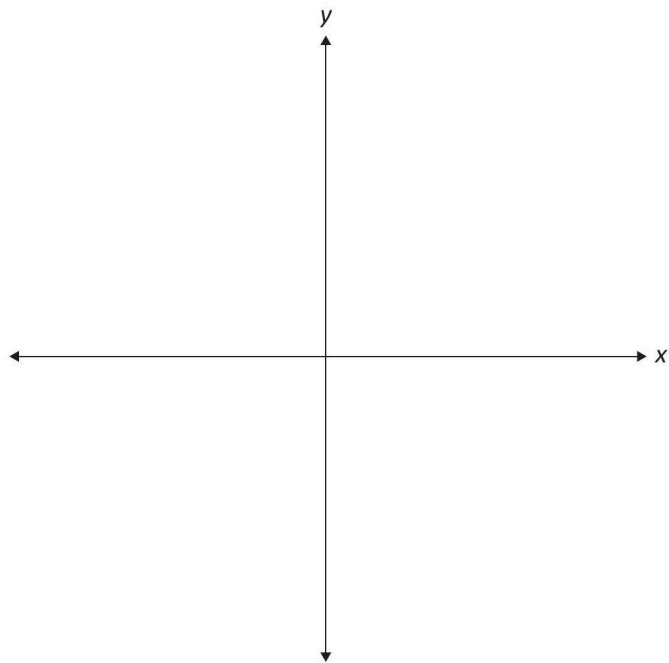
1 mark for vertical translation

2 marks

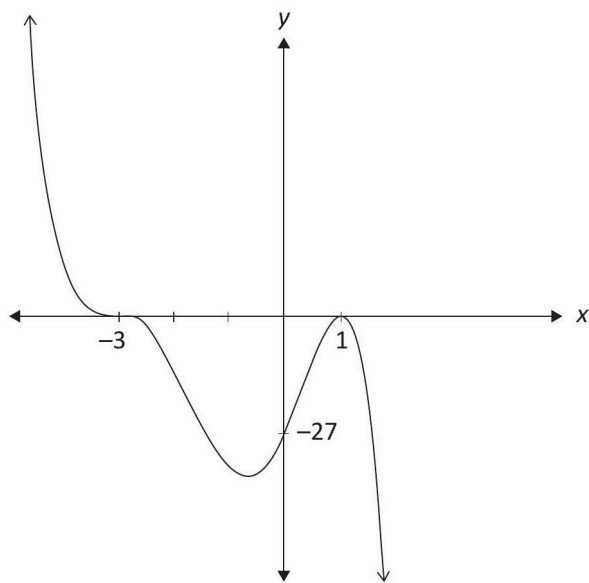
Question 15

Sketch the graph of a polynomial function, $p(x)$, with the following characteristics:

- degree 5
- leading coefficient of -1
- a zero at -3 , with a multiplicity of 3
- a zero at 1 , with a multiplicity of 2



Solution



1 mark for x -intercept at -3

with a multiplicity of 3

1 mark for x -intercept at 1

with a multiplicity of 2

$\frac{1}{2}$ mark for end behaviour

$\frac{1}{2}$ mark for y -intercept

3 marks

Question 16

Identify the expression that is equivalent to $\tan \theta (\csc^2 \theta - 1)$.

A.

$$\cot \theta$$

B.

$$\tan \theta$$

C.

$$\sec \theta$$

D.

$$\sin \theta$$

Solution

Correct answer: A

$$\cot \theta$$

Question 17

Identify the expression that represents the number of ways five books can be arranged on a shelf if two of them must be together.

A.

$$5!2!$$

B.

$$5! - 2!$$

C.

$$4!2!$$

D.

$$3!2!$$

Solution

Correct answer: C

$$4!2!$$

Question 18

Identify the value of x in the equation, $\log_8 x = -\frac{1}{3}$.

A.

$$-2$$

B.

$$-\frac{1}{2}$$

C.

$$\frac{1}{2}$$

D.

$$2$$

Solution

Correct answer: C

$$\frac{1}{2}$$

Question 19

Given the functions, $f(x) = \frac{1}{x}$ and $g(x) = \sqrt{x-1}$, identify the domain of $f(g(x))$.

A.

$$(-\infty, \infty)$$

B.

$$(-\infty, 1) \cup (1, \infty)$$

C.

$$[1, \infty)$$

D.

$$(1, \infty)$$

Solution

Correct answer: D

$$(1, \infty)$$

Question 20

Given $f(x) = \frac{3(x+2)}{5(x+2)(x-2)}$, identify the equation of the horizontal asymptote.

A.

$$y = -2$$

B.

$$y = 0$$

C.

$$y = \frac{3}{5}$$

D.

$$y = 2$$

Solution

Correct answer: B

$$y = 0$$

Question 21

Identify the range of the sinusoidal function, $f(x) = 2 \sin x + 3$.

A.

$$\{y \mid y \in \mathbb{R}\}$$

B.

$$\{y \mid 2 \leq y \leq 4, y \in \mathbb{R}\}$$

C.

$$\{y \mid -2 \leq y \leq 2, y \in \mathbb{R}\}$$

D.

$$\{y \mid 1 \leq y \leq 5, y \in \mathbb{R}\}$$

Solution

Correct answer: D

$$\{y \mid 1 \leq y \leq 5, y \in \mathbb{R}\}$$

Question 22

Given that $(x - 3)$ is a factor of the polynomial $P(x)$, identify which of the following is true.

A.

$$P(3) = 0$$

B.

$$P(0) = 3$$

C.

$$P(0) = -3$$

D.

$$P(-3) = 0$$

Solution

Correct answer: A

$$P(3) = 0$$

Question 23

Identify the value that best represents the angle in standard position.

A.

−4.4 radians

B.

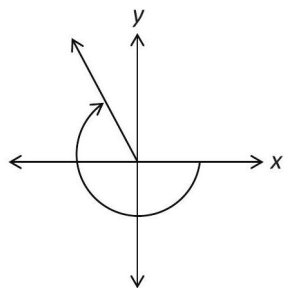
• 2 radians

C.

1.3 radians

D.

1.9 radians



Solution

Correct answer: A

−4.4 radians

Question 24

Identify the expression that is equivalent to $\log_a(x+2) + 3\log_a(x)$.

A.

$$\log_a(3(x+2))$$

B.

$$\log_a(x^3(x+2))$$

C.

$$\log_a\left(\frac{(x+2)}{x^3}\right)$$

D.

$$\log_a x + \log_a 2 + \log_a x^3$$

Solution

Correct answer: B

$$\log_a(x^3(x+2))$$

Question 25

Evaluate.

$$\tan^3\left(\frac{3\pi}{4}\right) + \csc\left(\frac{-4\pi}{3}\right) \cos\left(\frac{13\pi}{6}\right)$$

Solution

$$(-1)^3 + \left(\frac{2}{\sqrt{3}}\right)\left(\frac{\sqrt{3}}{2}\right)$$

$$-1 + 1$$

$$0$$

1 mark for $\tan\left(\frac{3\pi}{4}\right)$ (1/2 mark for value, 1/2 mark for quadrant)

1 mark for $\csc\left(\frac{-4\pi}{3}\right)$ ($\frac{1}{2}$ mark for value, $\frac{1}{2}$ mark for quadrant)

1 mark for $\cos\left(\frac{13\pi}{6}\right)$ ($\frac{1}{2}$ mark for value, $\frac{1}{2}$ mark for quadrant)

3 marks

Question 26

If $a = \log 4$ and $b = \log 3$, express $\log 36$ in terms of a and b .

Solution

$$\begin{aligned}\log 36 &= \log 4 + \log 3^2 \\ &= \log 4 + 2\log 3 \\ &= a + 2b\end{aligned}$$

1 mark for product law

1 mark for power law

2 marks

Question 27

Justify that $\tan 2x$ has a non-permissible value at $x = \frac{\pi}{4}$.

Solution

Method 1

$$\tan 2x$$

$$\text{Let } x = \frac{\pi}{4},$$

$$\tan\left(2 \cdot \frac{\pi}{4}\right) \quad 1 \text{ mark for substitution}$$

$$\tan \frac{\pi}{2}$$

$$\frac{1}{0} \text{ which is undefined}$$

1 mark for justification

$$\therefore x \neq \frac{\pi}{4}$$

2 marks

Method 2

$$\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$$

$$1 - \tan^2 x \neq 0$$

$$\tan^2 x \neq 1$$

$$\tan x \neq \pm 1$$

$$\therefore x \neq \frac{\pi}{4}$$

1 mark for correct substitution of appropriate identity

1 mark for justification

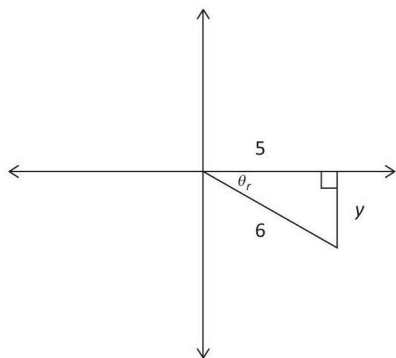
2 marks

Question 28

Determine the exact value of $\tan \theta$, if $\sec \theta = \frac{6}{5}$ and θ terminates in quadrant IV.

Solution

Method 1



$$\cos \theta = \frac{5}{6}$$

$$5^2 + y^2 = 6^2$$

$$y^2 = 36 - 25$$

$$y^2 = 11$$

1 mark for value of y

1 mark for $\tan \theta$ ($\frac{1}{2}$ mark for quadrant, $\frac{1}{2}$ mark for value)

2 marks

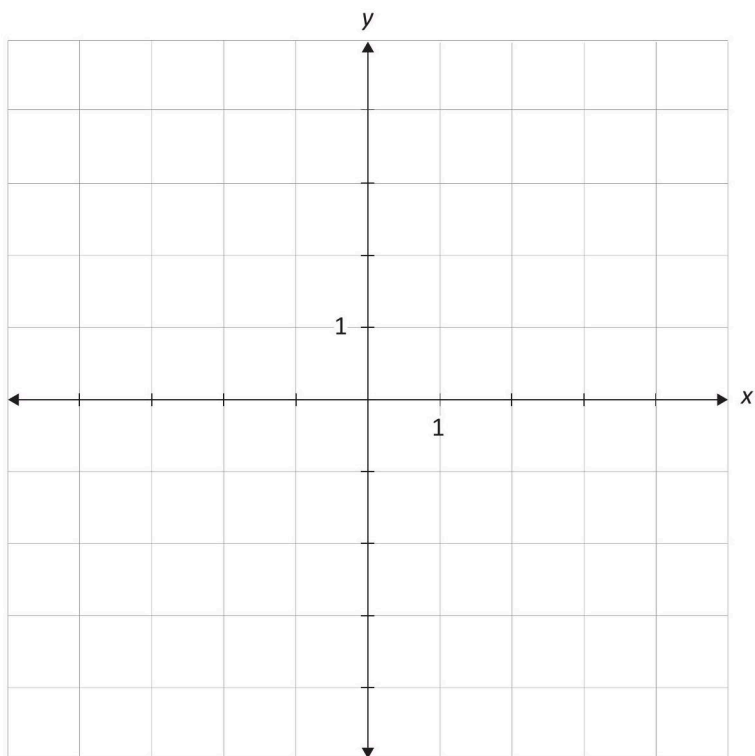
Note:

Accept any of the following values for y : $y = \pm\sqrt{11}$, $y = \sqrt{11}$, $y = -\sqrt{11}$.

Question 29

Given the function, $g(x) = \frac{4}{x^2+1}$,

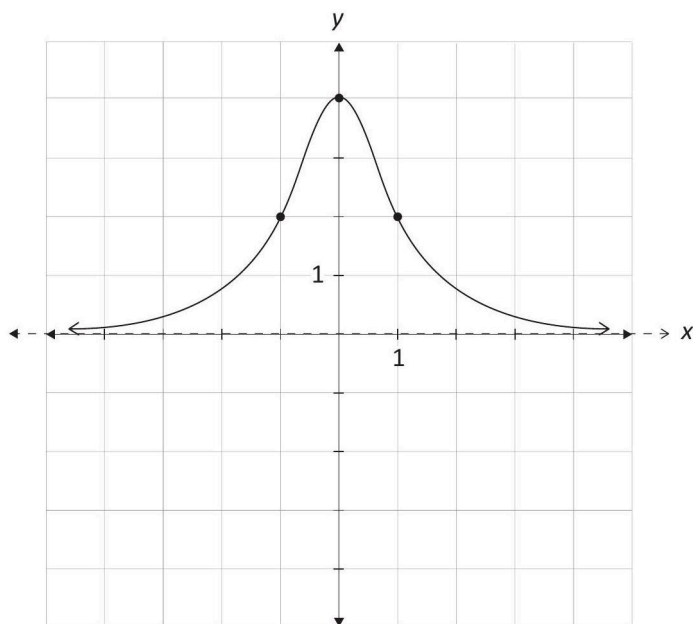
a) sketch the graph of $g(x)$.



b) state the range of the graph of $g(x)$.

Solution

a)



1 mark for asymptotic

behaviour approaching

$$y = 0$$

1 mark for shape

($\frac{1}{2}$ mark for graph left of $x = 0$;

$\frac{1}{2}$ mark for graph right of $x = 0$)

2 marks

b) Range: $\{y \mid 0 < y \leq 4, y \in \mathbb{R}\}$

or

Range: $(0,4]$

1 mark for range

consistent with a)

1 mark

Question 30

Given $y = x^2 - 4$,

a) determine the equation of the inverse.

$y =$

b) state a restriction on the domain of $y = x^2 - 4$, in order for its inverse to be a function.

Solution

a)

$$y = x^2 - 4$$

$$x = y^2 - 4$$

$$y^2 = x + 4$$

$$y = \pm\sqrt{x + 4}$$

1 mark for switching x and y

1 mark for solving for y

2 marks

b) Domain: $\{x \mid x \geq 0, x \in \mathbb{R}\}$

or

1 mark

Domain: $[0, \infty)$

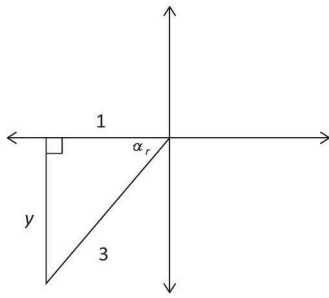
Note:

Other solutions are possible for b).

Question 31

Given $\cos \alpha = -\frac{1}{3}$ and $\sin \beta = -\frac{2}{3}$ where α and β terminate in the same quadrant, determine the exact value of $\sin(\alpha - \beta)$.

Solution



$$\xleftrightarrow{c} x^{\beta_r}_3$$

$$x^2 + y^2 = r^2$$

$$1 + y^2 = 9$$

$$y^2 = 8$$

$$y = \pm\sqrt{8}$$

$$y = -\sqrt{8}$$

$$x^2 + y^2 = r^2$$

$$x^2 + 4 = 9$$

$$x^2 = 5$$

$$x = \pm\sqrt{5}$$

$$x = -\sqrt{5}$$

$\frac{1}{2}$ mark for value of x

$\frac{1}{2}$ mark for value of y

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$= \left(\frac{-\sqrt{8}}{3} \right) \left(\frac{-\sqrt{5}}{3} \right) - \left(\frac{-1}{3} \right) \left(\frac{-2}{3} \right)$$

$$= \frac{\sqrt{40}}{9} - \frac{2}{9}$$

$$= \frac{\sqrt{40} - 2}{9}$$

or

$$= \frac{2\sqrt{10} - 2}{9}$$

$\frac{1}{2}$ mark for consistent value of $\sin \alpha$

$\frac{1}{2}$ mark for consistent value of $\cos \beta$

1 mark for substitution into correct

identity

3 marks

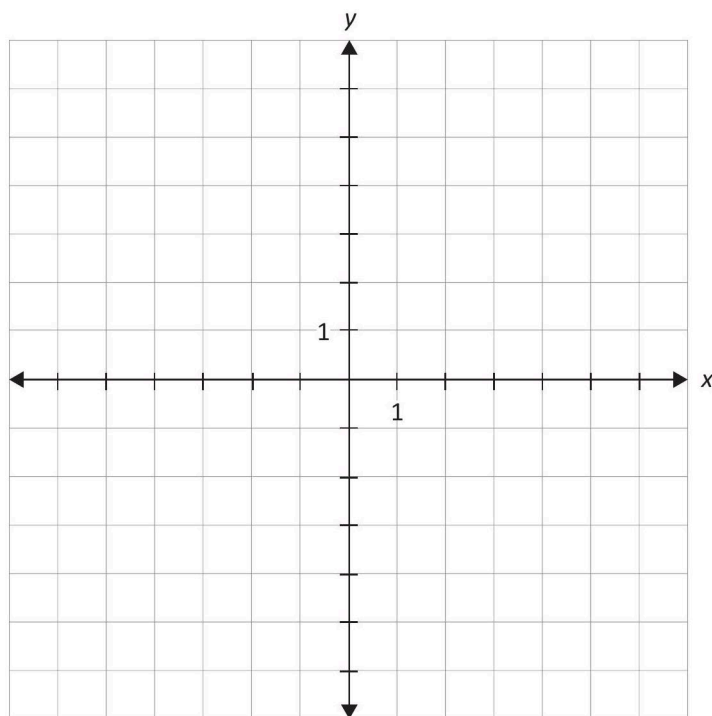
Note:

Accept any of the following values for x : $x = \pm\sqrt{5}, x = -\sqrt{5}, x = \sqrt{5}$.

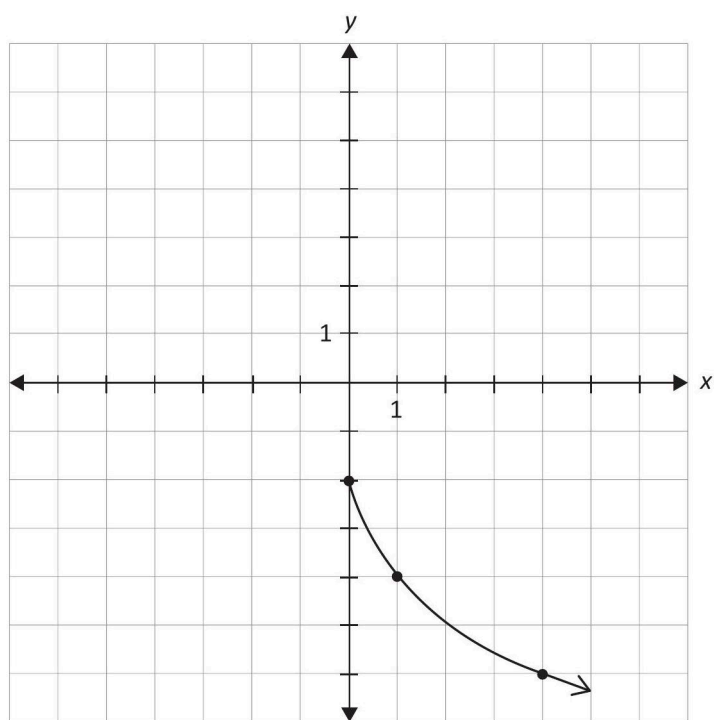
Accept any of the following values for y : $y = \pm\sqrt{8}, y = -\sqrt{8}, y = \sqrt{8}$.

Question 32

Sketch the graph of $y + 2 = -\sqrt{4x}$.



Solution



1 mark for shape of

a radical function

1 mark for vertical

reflection

1 mark for horizontal

compression

1 mark for vertical

translation

4 marks

Question 33

Express $p(x) = x^4 - x^3 - 6x^2 + 4x + 8$ in completely factored form, given that $(x + 1)$ is a factor of $p(x)$.

$p(x) =$

Solution

-1	1	-1	-6	4	8
		-1	2	4	-8
	1	-2	-4	8	0

$\frac{1}{2}$ mark for synthetic division

(or equivalent strategy)

$$p(x) = (x + 1)(x^3 - 2x^2 - 4x + 8)$$
$$p(2) = (2 + 1)(2^3 - 2(2)^2 - 4(2) + 8)$$
$$p(2) = 0$$

1 mark for identifying a second zero

of $p(x)$

$\therefore (x - 2)$ is a factor

2	1	-2	-4	8
		2	0	-8
	1	0	-4	0

$\frac{1}{2}$ mark for synthetic division

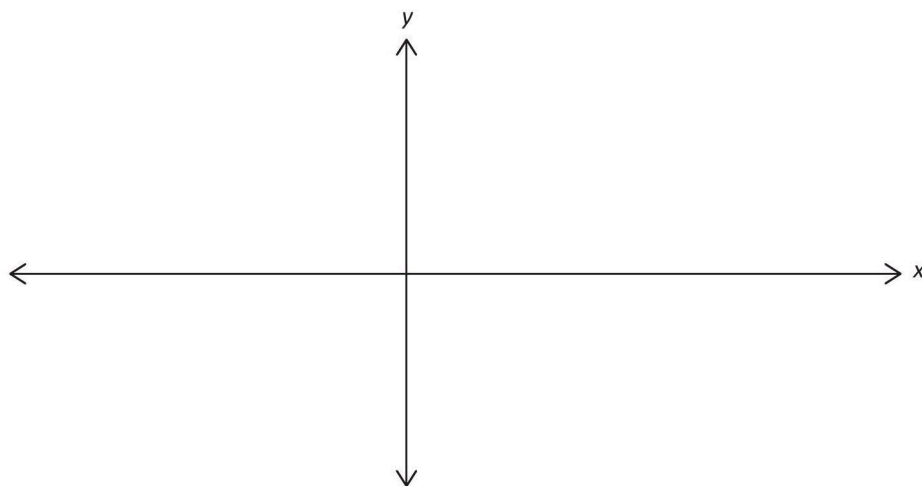
(or equivalent strategy)

1 mark for consistent product of factors

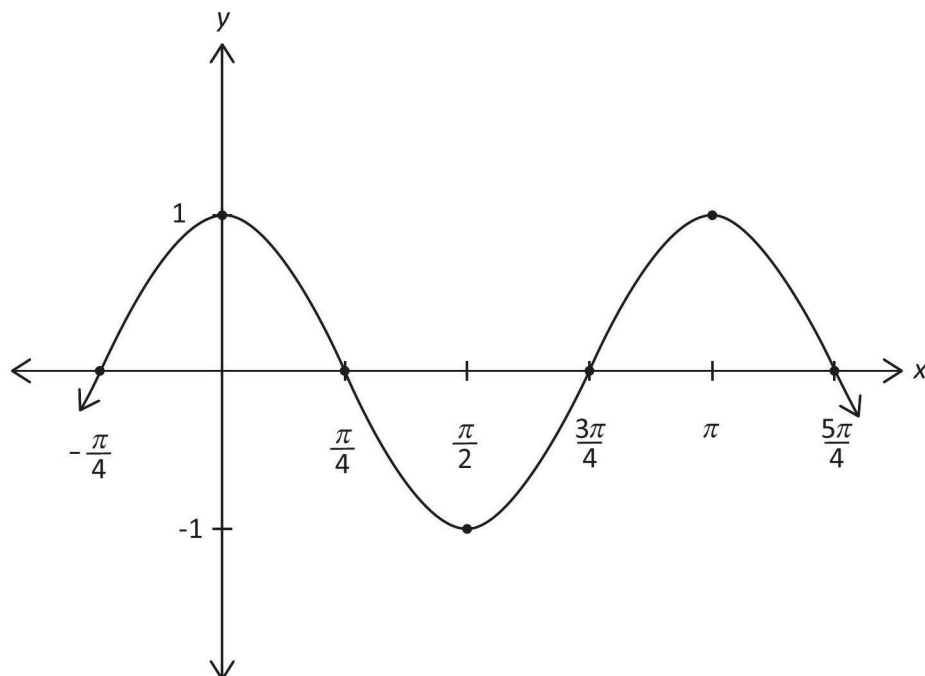
3 marks

Question 34

Sketch at least one period of the graph of $y = -\sin\left(2\left(x - \frac{\pi}{4}\right)\right)$.



Solution



1 mark for shape of a sinusoidal function with correct amplitude

1 mark for vertical reflection

1 mark for horizontal translation

1 mark for period

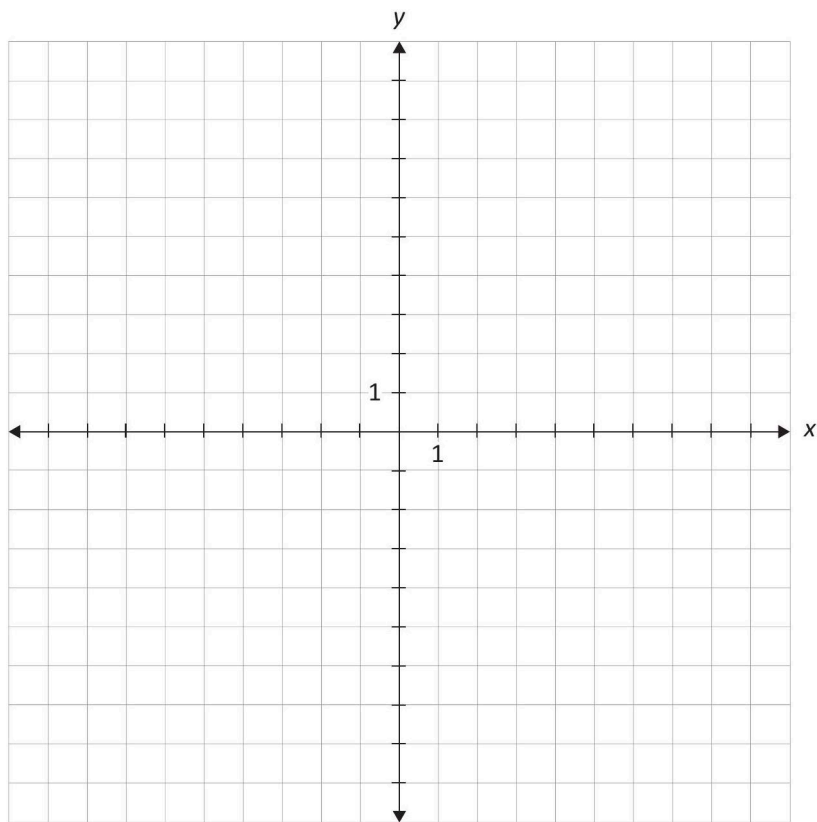
4 marks

Question 35

Given $g(x) = \log_2(x) - 3$,

a) determine the x -intercept of the graph.

b) sketch the graph of $g(x)$.



Solution

a)

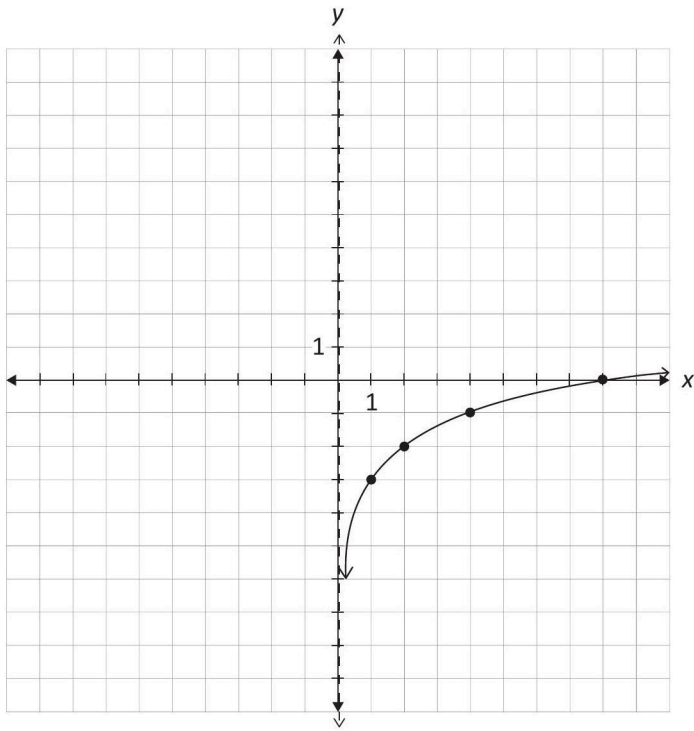
$$0 = \log_2(x) - 3$$

$$3 = \log_2(x)$$

$$2^3 = x$$

$$x = 8$$

b)



1 mark

1 mark for shape of
increasing logarithmic
function with base 2

1 mark for vertical
translation

2 marks

Question 36

Modesola incorrectly evaluated the expression, $\cos^2(30^\circ) - \sin^2(30^\circ)$.

Modesola's answer:

$$\begin{aligned}\cos^2(30^\circ) - \sin^2(30^\circ) \\&= 2 \cos(30^\circ) \\&= 2 \left(\frac{\sqrt{3}}{2} \right) \\&= \sqrt{3}\end{aligned}$$

Describe their error.

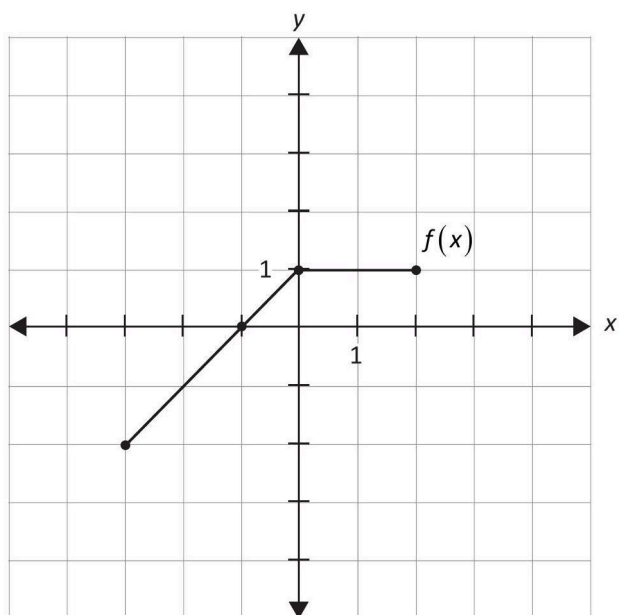
Solution

Modesola incorrectly applied the double angle identity. They should have written $\cos 2(30^\circ)$ instead of $2 \cos(30^\circ)$.

1 mark

Question 37

Given the graph of $y = f(x)$, state the range of $y = \sqrt{f(x)}$.



Solution

$$\{y \mid 0 \leq y \leq 1, y \in \mathbb{R}\}$$

or

$$[0, 1]$$

1 mark

Question 38

Determine a value of x that satisfies $\sin^2(2x - 1) + \cos^2(x + 5) = 1$.

Solution

$$2x - 1 = x + 5$$

$$x = 6$$

1 mark

Question 39

Solve, algebraically.

$$9^{2x+3} = \left(\frac{1}{3}\right)^{x+1}$$

Solution

$$(3^2)^{2x+3} = (3^{-1})^{x+1} \quad 1 \text{ mark for changing to a common base}$$

$$3^{4x+6} = 3^{-x-1}$$

$$4x + 6 = -x - 1 \quad 1 \text{ mark for equating exponents with a common base}$$

$$5x = -7 \quad 2 \text{ marks}$$

$$x = -\frac{7}{5}$$

Question 40

Determine the total number of arrangements of the letters in the word GUIDES if the vowels and consonants must alternate.

Solution

Method 1

$$\underline{6} \cdot \underline{3} \cdot \underline{2} \cdot \underline{2} \cdot \underline{1} \cdot \underline{1} = 72$$

1 mark for beginning with any letter

1 mark for alternating consonants and vowels

2 marks

Method 2

$$\text{case 1: } \frac{3}{v} \cdot \frac{3}{c} \cdot \frac{2}{v} \cdot \frac{2}{c} \cdot \frac{1}{v} \cdot \frac{1}{c} = 36$$

$$\text{case 2: } \frac{3}{c} \cdot \frac{3}{v} \cdot \frac{2}{c} \cdot \frac{2}{v} \cdot \frac{1}{c} \cdot \frac{1}{v} = 36$$

1 mark for alternating consonants and vowels

$\frac{1}{2}$ mark for second case

$\frac{1}{2}$ mark for addition of cases

2 marks

Question 41

Solve, algebraically.

$$\frac{5!(n-5)!}{3!(n-6)!} = 80$$

Solution

$$\frac{5 \cdot 4 \cdot 3!(n-5)(n-6)!}{3!(n-6)!} = 80$$

$$20(n-5) = 80$$

$$n-5 = 4$$

$$n = 9$$

1 mark for factorial expansion ($\frac{1}{2}$ mark for each)

$\frac{1}{2}$ mark for simplification of factorials

$\frac{1}{2}$ mark for solving for n

2 marks

Question 42

Justify that 2.1 is a better estimate than 2.6 for the value of $\log_5 29$.

Solution

$$5^2 = 25$$

$$5^3 = 125$$

$$\log_5 25 = 2$$

$$\log_5 125 = 3$$

29 is closer to 25 than 125

$\therefore \log_5 29$ is closer to 2 than 3

1 mark

Question 43

When $p(x) = x^3 + kx + 6$ is divided by $(x + 2)$, the remainder is 4 . Determine the value of k .

Solution

Method 1

$$4 = (-2)^3 + k(-2) + 6$$
$$4 = -8 - 2k + 6$$
$$4 = -2 - 2k$$
$$6 = -2k$$
$$k = -3$$

$\frac{1}{2}$ mark for $x = -2$

1 mark for remainder theorem

$\frac{1}{2}$ mark for solving for k

2 marks

Method 2

$$x = -2$$

$\frac{1}{2}$ mark for $x = -2$

-2	1	0	k	6
		-2	4	$-2k - 8$
	1	-2	$k + 4$	$-2k - 2$

1 mark for synthetic division (or equivalent strategy)

$$-2k - 2 = 4$$
$$-2k = 6$$
$$k = -3$$

$\frac{1}{2}$ mark for solving for k

2 marks

Question 44

Describe the transformations used to obtain the graph of the function $y = -f(7x + 14)$ from the graph of $y = f(x)$.

Solution

- reflection over the x -axis
- horizontal stretch by a factor of $\frac{1}{7}$
- horizontal translation 2 units to the left

1 mark for vertical reflection

1 mark for horizontal stretch

1 mark for horizontal translation

3 marks

Note:

Deduct a maximum of 1 mark for concept error of incorrect order of horizontal transformations.